

Python Programming Fundamentals

Why Dictionaries?

Programming Fundamentals Team

SETU

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Outline

1. Lists Are Good, But...
2. Dictionaries Use Keys
3. Records
4. Lists of Dictionaries



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A list can store related values:

```
student = ["Ava", 19, "Computing", 82]
```



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But what does each position mean?



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2. Dictionaries Use Keys

A dictionary stores data as key-value pairs.

```
student = {
    "name": "Ava",
    "age": 19,
    "course": "Computing",
    "mark": 82
}
```



2. Dictionaries Use Keys

A dictionary stores data as key-value pairs.

```
student = {
    "name": "Ava",
    "age": 19,
    "course": "Computing",
    "mark": 82
}
```

To access a value, we use its key:

```
print(student["course"]) # Computing
```



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3. Records

A dictionary is a natural way to represent a **record**.

Examples:

- one student
- one book
- one dog
- one product
- one planet



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This prepares students for:

- JSON
- APIs
- file storage
- CRUD applications



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4. Lists of Dictionaries

A list can store many dictionary records.

```
students = [
    {"name": "Ava", "mark": 82},
    {"name": "Ben", "mark": 71},
    {"name": "Cara", "mark": 90}
]
```



4. Lists of Dictionaries

A list can store many dictionary records.

```
students = [
    {"name": "Ava", "mark": 82},
    {"name": "Ben", "mark": 71},
    {"name": "Cara", "mark": 90}
]
```

This is our first simple database structure.



Thanks for Watching - Any questions?

